

Lab Manual: Exploring Streamlit Core Components

Objective

By the end of this lab, you will:

- Use **Markdown formatting** inside Streamlit apps.
- Work with **input widgets** like text fields, numbers, checkboxes, buttons, and dropdowns.
- Organize your app using **layouts and columns**.
- Display outputs such as tables and text dynamically.

Prerequisites

- Python 3.7 or higher installed.
- Streamlit installed: `pip install streamlit`
- A working `app.py` file from the previous lab (or create a new one).

Step 1: Create a New Streamlit App

```
import streamlit as st
import pandas as pd

st.title("Core Components & Layout")
```

Explanation:

- `import streamlit as st` → loads the Streamlit library.
- `import pandas as pd` → loads Pandas for data handling.
- `st.title("...")` → displays a large title at the top of the page.

Step 2: Add Markdown Text

```
st.markdown("This is bold text, this is italic text, and here is  
a [link](https://streamlit.io).")
```

Explanation:

- `st.markdown(...)` → lets you format text using Markdown.
- `bold` makes text bold.
- `italic` makes text italic.
- `[link](URL)` creates a clickable hyperlink.

Step 3: Add Basic Input Widgets

```
# Text input  
name = st.text_input("Enter your name:")  
  
# Number input with range  
age = st.number_input("Enter your age:", min_value=0, max_value=120,  
step=1)  
  
# Checkbox  
subscribe = st.checkbox("Subscribe to newsletter")  
  
# Button  
if st.button("Submit"):  
    st.write(f"Name: {name}, Age: {age}, Subscribed: {subscribe}")
```

Explanation:

- `st.text_input(...)` → creates a textbox where users type text.
- `st.number_input(..., min_value, max_value, step)` → creates a number input field with restrictions.
- `st.checkbox(...)` → returns True if checked, False if unchecked.
- `st.button("Submit")` → creates a clickable button. When clicked, the code inside `if` runs.

- `st.write(...)` → displays the results dynamically on the page.

Step 4: Use a Dropdown (Selectbox)

```
color = st.selectbox("Choose a color:", ["Red", "Green", "Blue"])
st.write(f"You selected: {color}")
```

Explanation:

- `st.selectbox("label", options)` → lets users pick one option from a dropdown list.
- The chosen value is stored in the variable `color`.
- `st.write(...)` displays the selected choice.

Step 5: Create Layout with Columns

```
# Split the page into two columns
col1, col2 = st.columns(2)
```

```
with col1:
```

```
    st.write("Column 1: Slider Example")
    slider_value = st.slider("Select a value", 0, 100, 50)
    st.write(f"Slider value: {slider_value}")
```

```
with col2:
```

```
    st.write("Column 2: Radio Button Example")
    choice = st.radio("Pick one:", ["Option A", "Option B", "Option C"])
    st.write(f"You chose: {choice}")
```

Explanation:

- `st.columns(2)` → splits the screen into **two equal-width columns**.
- `with col1:` → places widgets inside the first column.
 - `st.slider("label", min, max, default)` → creates a slider input.
- `with col2:` → places widgets inside the second column.

- `st.radio("label", options)` → lets users choose one option via radio buttons.

Step 6: Display Tabular Data

```
# Create a sample DataFrame
data = pd.DataFrame({
    "Fruit": ["Apple", "Banana", "Orange"],
    "Quantity": [10, 20, 15]
})
```

```
# Display as a static table
st.table(data)
```

Explanation:

- `pd.DataFrame({...})` → creates a simple dataset with fruit names and quantities.
- `st.table(data)` → displays it as a **static table**.
 - Unlike `st.dataframe()`, this table is not scrollable or interactive.

Step 7: Run the App

In your terminal: `streamlit run app.py`

- Streamlit starts a local web server.
- By default, the app runs at <http://localhost:8501> in your browser.

Verification Checklist

- Title and markdown text displayed.
- Text input, number input, checkbox, and button work correctly.
- Dropdown selection displays chosen value.
- Columns correctly display slider and radio button.
- DataFrame shows as a table.

Summary

In this lab, you:

- Learned to format text using **Markdown**.
- Collected input using **widgets** (text, number, checkbox, button, selectbox).
- Organized layouts with **columns**.
- Displayed tabular data using **st.table**.

These are the **core building blocks** for interactive Streamlit apps. With them, you can create dashboards, forms, and interactive data tools that feel polished and professional.